DATA SCIENCE SYLLABUS



MODULE 1: PYTHON & PROGRAMMING FOUNDATIONS

- Introduction to Python, installation and development environments
- Variables, data types, operators and type conversion
- Input/output and conditional statements
- Loops: for, while, break and continue
- Strings and string manipulation
- Data structures: Lists, Tuples, Sets and Dictionaries
- Functions, lambda functions, recursion and list comprehensions
- Modules, packages and virtual environments
- File handling: TXT, CSV, JSON and Excel files
- Exception handling and debugging
- Object-Oriented Programming: Classes, Objects, Inheritance, Encapsulation and Polymorphism
- Python DSA fundamentals: Searching, Sorting, Stacks, Queues and basic Linked Lists
- Git and GitHub: Repositories, commits, branching and version control
- Practical Work: Python programming exercises, file-processing applications and coding assignments.

MODULE 2: MATHEMATICS & STATISTICS FOR DATA SCIENCE

- Descriptive statistics: Mean, Median, Mode, Range, Quartiles and Percentiles
- Measures of dispersion: Variance, Standard Deviation and Interquartile Range
- Probability fundamentals and conditional probability
- Bayes' Theorem
- Probability distributions: Normal, Binomial and Poisson
- Sampling methods and sampling bias
- Inferential statistics and confidence intervals
- Hypothesis testing, p-values, Z-tests, T-tests and ANOVA
- Correlation, covariance and regression fundamentals
- Linear algebra: Vectors, Matrices, Matrix Multiplication and Dot Products
- Eigenvalues and Eigenvectors: Introduction
- Calculus fundamentals: Derivatives, Partial Derivatives and Gradients
- Gradient Descent and its role in machine learning
- Practical Work: Statistical analysis using Python, NumPy, Pandas and SciPy.

MODULE 3: DATABASE MANAGEMENT & SQL MASTERY

- Introduction to databases and RDBMS
- Tables, records, data types and database constraints
- Primary Keys, Foreign Keys and database relationships
- SQL commands: DDL, DML, DQL and transaction basics
- SELECT, WHERE, DISTINCT, ORDER BY and LIMIT
- Aggregate functions: COUNT, SUM, AVG, MIN and MAX
- GROUP BY, HAVING and CASE statements
- Joins: INNER, LEFT, RIGHT, FULL OUTER and SELF JOIN
- Subqueries and Common Table Expressions (CTEs)
- Temporary tables, views and stored procedures
- Window functions: ROW_NUMBER(), RANK(), DENSE_RANK(), LEAD() and LAG()
- Database normalization: 1NF, 2NF and 3NF
- Indexes and basic query optimization
- Hands-on practice using MySQL or PostgreSQL

- Practical Work: Customer segmentation queries, sales analysis and database reporting.

MODULE 4: DATA ANALYSIS, EDA & VISUALIZATION

NumPy and Pandas

- NumPy arrays, indexing, slicing and broadcasting
- Pandas Series and DataFrames
- Importing CSV, Excel, JSON and database data
- Data selection, filtering, sorting and aggregation
- Merging, joining, grouping and pivot tables
- Handling missing values, duplicate records and inconsistent data

Data Preprocessing

- Outlier detection using IQR and Z-score
- Feature scaling: Min-Max Scaling and Standardization
- Categorical encoding: One-Hot and Label Encoding
- Data transformation and data quality checks
- Exploratory Data Analysis (EDA)

Data Visualization

- Matplotlib: Line charts, bar charts, histograms and scatter plots
- Seaborn: Heatmaps, box plots, distribution plots and pair plots
- Correlation analysis and feature relationships
- Selecting suitable visualizations for business problems
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Business Intelligence

- Introduction to Power BI or Tableau
- Connecting data sources and preparing data
- Creating interactive dashboards, filters and drill-down reports
- Introduction to DAX in Power BI or calculated fields in Tableau
- Practical Work: Complete EDA on a real-world dataset and create a business dashboard.

MODULE 5: MACHINE LEARNING — SUPERVISED & UNSUPERVISED LEARNING

Machine Learning Fundamentals

- Introduction to Artificial Intelligence, Machine Learning and Deep Learning
- Types of machine learning: Supervised, Unsupervised and Reinforcement Learning
- ML workflow and problem definition
- Feature selection, feature engineering and preprocessing pipelines
- Training, validation and testing datasets
- Data leakage and overfitting prevention

Regression Algorithms

- Simple and Multiple Linear Regression
- Polynomial Regression
- Ridge and Lasso Regression
- Regression evaluation: MAE, MSE, RMSE and R^2

3. Classification Algorithms

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Naïve Bayes
- Support Vector Machines (SVM)
- Classification using Decision Trees

4. Tree-Based Models and Ensemble Learning
- Random Forest
- Bagging and Boosting
- Gradient Boosting
- Introduction to XGBoost, LightGBM and CatBoost

5. Unsupervised Learning
- K-Means Clustering
- Hierarchical Clustering
- DBSCAN
- Principal Component Analysis (PCA)
- Introduction to t-SNE for dimensionality reduction and visualization

6. Model Evaluation and Optimization
- Confusion Matrix, Accuracy, Precision, Recall and F1-Score
- ROC Curve and AUC
- Cross-validation and K-Fold validation
- Hyperparameter tuning using GridSearchCV and RandomizedSearchCV
- Class imbalance and appropriate resampling methods
- Scikit-learn pipelines and model comparison
- Introduction to time-series forecasting
- Practical Work: House price prediction, customer churn classification, customer segmentation and model comparison.

MODULE 6: DEEP LEARNING & COMPUTER VISION
- Introduction to deep learning and neural network architectures
- Artificial neurons, perceptrons and multilayer perceptrons
- Activation functions: ReLU, Sigmoid, Tanh and Softmax
- Forward propagation and backpropagation
- Loss functions and optimization algorithms
- Gradient Descent, SGD, Adam and RMSProp
- TensorFlow/Keras or PyTorch fundamentals
- Building and training Artificial Neural Networks (ANNs)
- Overfitting, Dropout, Batch Normalization and regularization
- Convolutional Neural Networks (CNNs)
- Convolution, filters, pooling and image classification
- Transfer learning using pretrained models such as VGG and ResNet
- Introduction to object detection
- Data augmentation and image preprocessing
- Model evaluation and performance improvement
- Practical Work: Image classification and transfer-learning projects using publicly available datasets.

MODULE 7: NATURAL LANGUAGE PROCESSING & GENERATIVE AI

Natural Language Processing

- Introduction to NLP and text analytics
- Text cleaning and preprocessing
- Tokenization, stop-word removal, stemming and lemmatization
- Regular expressions and text normalization
- Bag of Words (BoW) and TF-IDF
- Word embeddings: Word2Vec and GloVe
- Text classification and sentiment analysis
- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory (LSTM) networks

Modern NLP and Generative AI

- Attention mechanisms and Transformer architecture
- Introduction to Large Language Models (LLMs)
- Using pretrained models through Hugging Face Transformers
- Text generation, summarization and question answering
- Prompt engineering and responsible AI use
- Embeddings and semantic search
- Introduction to Retrieval-Augmented Generation (RAG)
- Building a basic document-question-answering application
- Introduction to vector databases and similarity search
- Fine-tuning concepts and parameter-efficient fine-tuning overview
- Practical Work: Sentiment analysis, text classification and a basic RAG-based document assistant.

MODULE 8: MODEL DEPLOYMENT & MLOps FUNDAMENTALS

- Saving and loading trained models using joblib or appropriate framework tools
- Building REST APIs using FastAPI or Flask
- Creating interactive applications using Streamlit
- Connecting deployed applications to trained models
- Input validation and error handling
- Git and GitHub for project management
- Basic testing and application documentation
- Introduction to Docker and containerization
- Model versioning and experiment tracking fundamentals
- Introduction to CI/CD workflows using GitHub Actions
- Model monitoring, data drift and model retraining concepts
- Introduction to cloud deployment options
- Practical Work: Deploy a machine learning prediction application with a simple web interface.

MODULE 9: CAPSTONE PROJECTS & CAREER PREPARATION

Project 1 — Finance

- Credit risk prediction
- Fraud detection using transaction data
- Risk scoring and classification evaluation

Project 2 — E-Commerce

- Customer churn prediction

- Customer segmentation
- Product recommendation system fundamentals
- Customer Lifetime Value (CLV) prediction

Project 3 — Healthcare

- Patient readmission risk prediction using suitable, appropriately de-identified data
- Clinical text classification using public datasets
- Model evaluation and responsible use of sensitive data

Project 4 — Computer Vision

- Image classification using CNNs
- Facial expression recognition
- Transfer learning and model performance comparison

Project 5 — NLP and Generative AI

- Sentiment analysis
- Document search and question answering using RAG
- AI-powered document assistant

Career Preparation and Placement Support

- Resume preparation and LinkedIn profile optimization
- GitHub portfolio review
- Python programming and coding interview practice
- SQL query challenges
- Statistics and machine learning interview questions
- Deep learning and NLP fundamentals revision
- Mock technical and HR interviews
- Project presentation and communication skills
- Career guidance for entry-level Data Analyst, Junior Data Scientist, ML Engineer Trainee and AI Developer Trainee roles

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